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Memory Game

Table of Contents:

Introduction: Page 2

Functional Requirements: Page 4

Nonfunctional Requirements: Page 5

System Requirements: Page 6

Future References: Page 9

Contributions: Page 9

Introduction:

This is a software project for a memory game. This will help older people with memory problems play an entertaining game that will help fight off short-term memory loss. The layout of the game will be a button memory game, where there will be a set amount of buttons, each unique, and they will play a sequence/order, and the user will have to repeat that order. Once the order has been successfully repeated, the buttons will play the same sequence that just occurred, but with an additional button click at the end.

The functionalities of the game will include:

1. The game itself, which is described in the paragraph above.
2. some form of difficulty setting, where there may be more buttons, or a certain goal that must be reached, or possibly how many buttons are incremented each sequence
3. a leaderboard section- this may be a local session, or an actual leaderboard, that saves between sessions.
4. A scoring mechanism- this would be how far a person gets, and would save to the leaderboard if they score high enough to place on it
5. A colorblind section- there are multiple forms of colorblindness, this would just make the buttons different colors
6. Settings, including text size, and sound volume- old people are playing this game and may not be able to read and hear too well, so this would help

The main languages of this software project will likely include HTML/CSS/JS, and will also likely use Git/Github. If there are sounds used this could use additional software, such as FL studio, or something simpler.

Potential users include first and foremost old people, which are the primary age of people who have memory issues, the latter of which this game is trying to combat. There could also be just people who like to play games in general/puzzle games. or those who want to test their short-term memory.

Benefits could include cognitive enhancement, entertainment and stress relief, and competition. Competition could be enhanced with the use of a leaderboard and may encourage motivation/competition with oneself.

Things that may have changed from the original paper requirements are that there likely will not be a time or moves counter, and there also has been added a colorblind ability. We have also chosen the main game's goal. We have also chosen the fact that HTML/CSS/JS will be the software of choice, compared to Python or Java. There also will likely be no need to use graphics implementation, which can be done in one of the other preexisting languages. A note is this game is subject to change, with the possible removal and addition of some features which may be big or small.

Requirements Specification:

Functional Requirements

0. There will be a main menu screen with buttons that take users to other screens: start button, settings button, difficulty button, leaderboard button
1. The game should have a start button to play the game.
2. The game should display the sequence of buttons for the player to remember, which increases in length every round
3. The player should be able to clearly replicate what sequence is generated on the computer end
4. The computer should check if the user's sequence is correct
5. If the user sequence was correct, the game should go to the next round
6. If the user sequence is not correct, the game will end, resulting in an end game screen
7. The game should keep track of the score and a possible goal the computer has set
8. The score should be displayed at the end screen, and possibly on the leaderboard if it reaches the criteria
9. The leaderboard shall display (3 or 10) of the current high scores, of that local session(or all time, if we have time to add a database)
10. The restart button, displayed on the end game screen, should restart the game to replay the game
11. The leaderboard button, displayed on the end game screen should properly display the leaderboard screen
12. The settings button will allow access to the settings screen
13. The settings screen should properly allow editing of the text size, volume, and other possible settings(colorblind mode)
14. The main menu screen should allow access to the difficulty screen via the difficulty button.
15. The difficulty screen should allow the user to change the number of buttons on the screen, and what game mode to play in(goal-oriented, or free play)
16. If the user is in goal-oriented mode, then once the user reaches the goal score, the goal screen should be displayed, and allow the user to restart, or possibly change the mode they are playing in some way

Nonfunctional Requirements

1. The game should have minimal delay, to less than one second when actually playing the game, and less than five seconds when switching between screens or settings
2. The game should be able to be run on almost any web browser
3. The game should provide a good user experience and be easy to follow
4. The code should have good practices, and be well organized, and maintainable.
5. The game should be accessible and easy to follow for old people or those who are not familiar with computers/screens

System requirements:

The Website Itself	What the user is using to interact with the game and program, should display the game for the user.
Inputs:	Inputs will come from the user of the website using a mouse or touchscreen. These will go to the game to decide what the output should be.
Outputs:	Outputs will be displayed on the website for the user to see and respond to. These will include things like results and feedback to the user from the game.
Action:	The website should display the game for the user to interact with and play and also give feedback to the user after receiving it from the game.

The Game	The program that is running sends feedback to the website that tells the website what to display.
Inputs:	Inputs will come from the website and will be interpreted and sent to functions in the program that is the game.
Outputs:	Outputs from the game/program will be the result of the match or the current score and also what to show on the website at the time. The output will be sent to the website in order to update it with what to show the user.
Action:	The action of the game is to process all the inputs and decide what the output should be based on the users' inputs. It also has to come up with the different sequences and check the users' answers. It will need to also keep track of the score and also the high score.

Sequence Creator/Checker	This will decide the sequence at random for the game and also store it in a data structure so the user's
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	answers can be checked against it. Must also be able to add new steps in the sequence.
Inputs:	The inputs will be from the website and will be what the user has pressed on the website in correlation to the game. These inputs will be sent to the function and used to do the checking of answers.
Outputs:	The Outputs will be the pattern that the program has come up with as well as the same pattern in a data structure. The output will also be whether or not the user's input matches the pattern that the function has given. This will be sent back to the website in order to display the proper information.
Action:	The function will either create a new pattern for the game, or the function will check the users' answers against the pattern that it has made and return the answer to the website for display.
Pre-Condition	The function will be called if the website wants to know if the user is correct and also will be called if the website needs a pattern for the user to memorize.
Post-Condition	There will be a pattern that is sent back to the website, or there will be a result sent back to the website based on the users' answers.

Scoreboard	This will keep track of both the current score and also the high scores (locally) and give the values to the website to keep track of.
Inputs:	The input of this function will be whether or not the user got the pattern right. These inputs will be used to either return the current score or add to the current score.
Outputs:	The output of this function will be the current score and also the high score of the game. These will be sent to the website to display for the user.
Action:	The function will use the input that it gets to either return the current score if the user got the answer wrong or the score plus the points that the user just got.
Pre-Condition	Function Will be called anytime the user makes an

	answer to either update the score for the website or to return the current score to the user.
Post-Condition	There will be a score sent to the website along with the high score.

List of References:

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Contributions:

Drew Kroeger: Introduction, Some user requirements

Kobe Essien- System Requirements

Owen Manley- Helped with organizing and creating user requirements

Matthew Farha- Got possible references, table of contents, overall review of paper